

Affordances are for life (and not just for maximizing reproductive fitness)

Edward Baggs¹, Guilherme Sanches de Oliveira, Miguel Segundo-Ortin, Paula Silva, Vicente Raja, Anthony Chemero

Abstract

We are pleased to see the concept of affordances being given attention in the context of evolutionary theory. We are, however, surprised that the authors engage so little with existing work on affordances. We highlight some properties of affordances that the authors overlook: affordances are perceivable, are ubiquitous across the animal kingdom, are relative to individuals, and can be learned.

Main text

We are grateful for the opportunity to comment on Ko & Neuberg's (K&N) thoughtful target article, which promotes the concept of affordances as having a valuable role to play within human evolutionary theory. We agree that affordances are valuable here, and we will take the opportunity to provide some theoretical context. The notion of affordances was introduced by James J. Gibson (1966, 1979), whom the authors acknowledge. Gibson's aim, which the authors do not acknowledge, was to develop a novel theory of perception based on the direct pickup of structure in ambient information. Affordances are central to the ecological approach to psychology, a tradition that builds on Gibson's work (Segundo-Ortin and Raja, 2024). We are surprised that the authors do not engage with the extensive literature in the ecological psychology tradition. We believe that doing so would allow the authors to address several weaknesses in their proposal. We will focus on four key points.

(1) *Affordances are perceivable*. Affordances, if they are to structure behavior, must be perceived. The target article fails to address the question of how perception is achieved for any of the cited examples. Empirical work in ecological psychology has sought to identify the precise ecological information implicated in the control of a given behavior (Wagman, 2020; Warren, 2021). For example, to safely control braking, the driver can attend to the global rate of optic flow that specifies their current rate of deceleration (Fajen, 2005; Gotardi et al., 2024). In the target article, K&N are interested in facts of a

¹ Correspondence: ebag@sdu.dk

different order, such as the genetic relatedness of potential partners, the ratio of men to women in the local population, or the possibility of alloparenting. An individual can gain knowledge about all these things, but it is unclear exactly what is meant to count as the affordance in these cases, far less how those affordances could be cashed out perceptually in terms of ecological information (Segundo-Ortin and Heras-Escribano, 2024). The knowledge in these cases is of a peculiarly human sort: it requires abstract rational thinking. This brings us to our second point.

(2) *Affordances are general to all animals.* In his later life, Charles Darwin carried out a series of empirical studies on earthworms (Darwin, 1881). Among other things, he discovered that these creatures selectively and efficiently plug the openings to their burrows by dragging in leaves by the narrow end first. As noted by Ed Reed, this is an “example of how behavior is regulated by affordances” (Reed, 1996, p. 21). If we accept Reed’s assessment, then we must conclude that affordances are prevalent across the animal kingdom (Wagman et al., 2019). K&N’s primary examples—avoiding inbreeding, violent environments, and unfavorable sex ratios—are relevant to all sexually reproducing animals, not just humans. But remember that affordances can affect behavior only to the extent that the animal is able to perceive the affordance in question. Can non-human animals perceive genetic relatedness? Interestingly, recent evidence suggests that many non-human species do not in fact have mechanisms to avoid inbreeding (de Boer et al., 2021; Pike et al., 2021). An important difference between humans and other animals is that we have symbolic culture, which enables us to conceive of the very notion of inbreeding and to create cultural taboos and customs against it. Apropos of which...

(3) *Affordances are relative to individuals, but humans live in groups!* The target article focuses on individuals’ ‘management’ of resources, through measures such as “relocating [one’s] family to access kin support” or hanging out in “e.g., clubs or bars on Friday nights” to find potential partners. The focus on individual behavior obscures the ways that K&N’s examples presuppose communal infrastructure such as, say, the peculiarly human and historically contingent notion of a ‘Friday night.’ By neglecting the communal and institutional dimensions of human behavior, K&N—perhaps inadvertently—sometimes couch socially rooted patterns as simple outcomes of human biology. For instance, persistent patterns of gender imbalance in parental and kin investment are linked to biological reproductive asymmetries such as the costs of gestation and genetic certainty. While such asymmetries exist, this focus neglects the contribution of social norms, cultural institutions, and economic structures in generating unequal caregiving responsibilities, and obscures the fact that caregiving arrangements are amenable to cultural change and collective reform. More troubling still is K&N’s discussion of sexual violence. They describe male violence against women—particularly young women of reproductive age—as a strategy to control female sexuality for reproductive advantage. This framing threatens to naturalize gender-based harm by treating it as an adaptive reproductive behavior of heterosexual males. Moreover, it portrays individual solutions to violence, like threat ‘management’ on the part of women, as desirable rather than highlighting the need for community-scale solutions that address its cultural, political, and institutional determinants (Chater and Loewenstein 2024).

Typically it is human communities, not individuals, that develop solutions to deal with genetic relatedness (marriage customs), physical violence (laws, punishments), and skewed sex ratios (government campaigns in many countries). Individual management of affordances alone cannot explain such community-scale mechanisms (Baggs, 2021; Heft, 2001) which are explicitly negotiated and require

human language and reasoning. Ultimately, a theory of affordances in humans must be accompanied by a theory of how children learn to become members of their community (Baggs and Chemero, 2020; Chemero, forthcoming). This is a process that takes time...

(4) *Affordances change continuously, not discretely, across development.* K&N inform us that “[l]ittle is known [...] about how human-environment interactions vary across the lifespan.” On the contrary, the study of development has been a key strand of the ecological program, from Eleanor Gibson’s classic studies of infants on the visual cliff (Gibson and Walk, 1960) and subsequent studies of infant and child learning (Adolph and Hoch, 2019; Thelen and Smith, 1994) to work on skill acquisition (Silva et al., 2019) and motor rehabilitation in adults (Silva and Schwab, 2024; Vaz et al., 2017). The target article discretizes human development around two life transitions: puberty and parenthood. In reality, the body continuously changes, and affordances therefore continuously change too. We invite the authors to explain how the affordance management framework deals with such continuous bodily changes.

Given these four points, we find that the authors are in danger of trying to reinvent several existing lines of research. There is an enormous published literature about affordance perception, and the target article would have benefited had K&N consulted it. We hope that the authors will use the works we cite here as guides into the literature on affordances.

Acknowledgments

We thank Britain’s Dogs Trust charity, formerly the National Canine Defence League, whose campaign slogan, “A dog is for life, not just for Christmas,” we have adapted for our title.

Funding statement

Guilherme Sanches de Oliveira is supported by the grant “Embodied Cognitive Science and Scientific Expertise in Reading” (22831/SF/24), funded by the ‘Saavedra Fajardo’ sub-programme of Fundación Séneca, the Science and Technology Agency of the Region of Murcia. Miguel Segundo-Ortin is supported by the grant “Transformando la ciencia cognitiva comparada: aplicaciones de la psicología ecológica y el enactivismo para superar el neurocentrismo y los antropo-sesgos” (22586/JLI/24), funded by FSRM/10.13039/100007801, and by a Ramón y Cajal Fellowship from Ministerio de Ciencia e Innovación del Gobierno de España (Award #RYC2021-031242-I) funded by MCIN/AEI/10.13039/501100011033 and by “European Union NextGenerationEU/PRTR”. Vicente Raja is supported by a Ramón y Cajal Fellowship from Ministerio de Ciencia e Innovación del Gobierno de España (Award #RYC2023-045124-I) funded by MCIN/AEI/10.13039/501100011033. Anthony Chemero is funded by the Charles Phelps Taft Research Center at the University of Cincinnati.

Conflicting interests

The authors declare no competing interests.

References

- Adolph, K. E. & Hoch, J. E. (2019). Motor development: Embodied, embedded, enculturated, and enabling. *Annual Review of Psychology*, 70(141–164).
- Baggs, E. (2021). All affordances are social: Foundations of a Gibsonian social ontology. *Ecological Psychology*, 33(3-4):257–278.

- Baggs, E. & Chemero, A. (2020). Thinking *with* other minds. *Behavioral and Brain Sciences*, 43(e90):24–25.
- Chater, N. & Loewenstein, G. (2023). The i-frame and the s-frame: How focusing on individual-level solutions has led behavioral public policy astray. *Behavioral and Brain Sciences*. 2023;46:e147.
- Chemero, A. (forthcoming). *Intertwined creatures: The embodied cognitive science of self and other*. Columbia University Press.
- Darwin, C. R. (1881). *The formation of vegetable mould through the action of earthworms*. John Murray.
- de Boer, R. A., Vega-Trejo, R., Kotrschal, A., and Fitzpatrick, J. L. (2021). Meta-analytic evidence that animals rarely avoid inbreeding. *Nature Ecology & Evolution*, 5(7):949–964.
- Fajen, B. R. (2005). Calibration, information, and control strategies for braking to avoid a collision. *Journal of Experimental Psychology: Human Perception and Performance*, 31(3):480–501.
- Gibson, E. J. and Walk, R. D. (1960). The “visual cliff”. *Scientific American*, 202(4):64–71.
- Gibson, J. J. (1966). *The Senses Considered as Perceptual Systems*. Houghton-Mifflin.
- Gibson, J. J. (1979). *The Ecological Approach to Visual Perception*. Houghton-Mifflin.
- Gotardi, G. C., Van der Kamp, J., Navarro, M., Savelsbergh, G. J., and Rodrigues, S. T. (2024). Affordance-based control of braking in cycling: Experience reveals differences in the style of control. *Human Movement Science*, 95:103225.
- Heft, H. (2001). *Ecological psychology in context: James Gibson, Roger Barker, and the legacy of William James's radical empiricism*. Lawrence Erlbaum Associates.
- Pike, V. L., Cornwallis, C. K., and Griffin, A. S. (2021). Why don't all animals avoid inbreeding? *Proceedings of the Royal Society B: Biological Sciences*, 288(1956):20211045.
- Reed E. S. (1996) *Encountering the world: Toward an ecological psychology*. Oxford University Press.
- Segundo-Ortin, M., & Heras-Escribano, M. (2024). The risk of trivializing affordances: mental and cognitive affordances examined. *Philosophical Psychology*, 37(7), 1639–1655.
- Segundo-Ortin, M., & Raja, V. (2024). *Ecological Psychology*. Cambridge University Press.
- Silva, P., Kiefer, A., Riley, M. A., & Chemero, A. (2019). Trading perception and action for complex cognition: Application of theoretical principles from ecological psychology to the design of interventions for skill learning. In M. L. Cappuccio (Ed.), *Handbook of embodied cognition and sport psychology* (pp. 47–74). The MIT Press.
- Silva, P. L., & Schwab, S. M. (2024). Disability through the Lens of affordances: a promising pathway for transforming physical therapy practice. In *The Modern Legacy of Gibson's Affordances for the Sciences of Organisms* (pp. 291–326). Routledge.
- Thelen, E. & Smith, L. B. (1994). *A dynamic systems approach to the development of cognition and action*. MIT Press.
- Vaz, D. V., Silva, P. L., Mancini, M. C., Carello, C., and Kinsella-Shaw, J. (2017) Towards an ecologically grounded functional practice in rehabilitation. *Human Movement Science*, 52: 117–132.
- Wagman, J. B. (2020). A guided tour of Gibson's theory of affordances. In *Perception as Information Detection: Reflections on Gibson's Ecological Approach to Visual Perception*, eds Wagman, J. B. and Blau, J. J. C. (New York: Routledge), 130–148.
- Wagman, J., Lozano, S., Covarrubias, P., Cabrera, F., & Jiménez, Á. (2019). Perception of affordances in the animal kingdom and beyond. In: I. Zepeda Riveros, F. Cabrera González, J. A. Camacho Candia and E. Camacho Gutiérrez (Eds.). *Aproximaciones al estudio del comportamiento y sus aplicaciones, Vol. II*. (pp. 70–108). Universidad de Guadalajara.
- Warren, W. H. (2021). Information is where you find it: perception as an ecologically well-posed problem. *I-Perception*, 12(2), 20416695211000366.

Author information

Edward Baggs

Department of Culture and Language, University of Southern Denmark, Odense, Denmark
Danish Institute for Advanced Study, University of Southern Denmark, Odense, Denmark
ebag@sdu.dk

Guilherme Sanches de Oliveira
Department of Philosophy, University of Murcia, Murcia, Spain
Minimal Intelligence Laboratory (MINT Lab), University of Murcia, Murcia, Spain
guilherme.sanches@um.es

Miguel Segundo-Ortin
Department of Philosophy, University of Murcia, Murcia, Spain
Minimal Intelligence Laboratory (MINT Lab), University of Murcia, Murcia, Spain
miguel.segundo@um.es

Paula Silva
Department of Psychology, University of Cincinnati, Cincinnati, OH, USA
silvapa@ucmail.uc.edu

Vicente Raja
Department of Philosophy, University of Murcia, Murcia, Spain
Minimal Intelligence Laboratory (MINT Lab), University of Murcia, Murcia, Spain
The Rotman Institute of Philosophy, University of Western Ontario, Ontario, Canada
vicente.raja@um.es

Anthony Chemero
Departments of Philosophy and Psychology, University of Cincinnati, Cincinnati, OH, USA
chemeray@ucmail.uc.edu